

Sandesh S. Kalantre

Knight-Hennessy Fellow, Stanford University | [Website](#) | sandykal@stanford.edu

EDUCATION

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- Stanford University**, GPA: 4.0/4.0 2022 - 2026 (*exp.*)
PhD in Physics, Knight-Hennessy Fellow
Advisor: Prof. David Goldhaber-Gordon
- University of Maryland (UMD), College Park**, GPA: 4.0/4.0 2018 - 2022
Lanzos Graduate Fellow (*transferred to Stanford*)
Advisor: Prof. James R. Williams
- Indian Institute of Technology (IIT), Bombay - India**, GPA: 9.87/10.0 2014- 2018
B.Tech in Engineering Physics with Honors and Minor in CS, Institute Silver Medal

RESEARCH SKILLS

Quantum Device Fabrication

Magnetron sputtering	E-beam evaporation	Atomic layer deposition	Photolithography
E-beam lithography	Reactive ion etching	AFM imaging	SEM imaging
2D materials	Josephson junctions	2D Stacking	μ -wave resonators

Cryogenic & Microwave Instrumentation

Dilution refrigerators	VTI	VNA spectroscopy	Quasi-DC transport
RF reflectometry	Custom mixer circuits	High-speed digitizers	Lock-in amplifiers

Simulation & Computational Tools

Python	C++	Qcodes	gdsapy/gdstk
NumPy	SciPy	QuTiP	TensorFlow
Sonnet	HFSS	COMSOL	SolidWorks
FreeCAD	Kicad	Metal machining	3D printing

PATENTS

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- ❖ Ray-based Classification Framework for Machine Learning-based Tuning Techniques 2023
J.P. Zwolak, **S.S. Kalantre**, J.M. Taylor, T.W. McJunkin, US Patent App. 18/028,154

RESEARCH HIGHLIGHTS

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- ❖ **Microwave spectroscopy of a thin film quantum spin liquid candidate.** Built a NbTiN superconducting resonator platform with flip-chip integration to measure spin resonance in epitaxial TbInO₃, opening a route to GHz-frequency probes of frustrated magnetism in films too small for conventional ESR. [Kalantre et. al.](#), under review at *Physical Review X*.
- ❖ **cQED architecture for superfluid stiffness in 2D superconductors.** Designed and validated a microwave platform for extracting superfluid stiffness in van der Waals superconductors at GHz frequencies and mK temperatures, applicable to rhombohedral graphene and other emergent 2D superconductors.
- ❖ **First ML-based autotuning of a quantum dot device.** Developed CNN classifiers and a closed-loop tuning algorithm that autonomously drove a semiconductor quantum dot to the single-electron regime in situ, establishing a new paradigm for automated quantum device control now adopted across the field. [Kalantre et. al.](#), *npj Quantum Information* (2019); US patent.

- ❖ **Resolved puzzling Shapiro measurements in driven graphene Josephson junctions.** Developed an analytical framework based on non-linear oscillator dynamics that explained anomalous Shapiro step structure in graphene Josephson junctions, reconciling microwave response measurements with junction parameters. [Kalantre et. al.](#), *Physical Review Research* (2020).

PUBLICATIONS

500+ citations, h-index = 10 ([Google Scholar](#))

13. **S.S. Kalantre**, J. Nordlander, M.A. Anderson, J.A. Mundy, and David Goldhaber-Gordon, Microwave spin resonance in epitaxial thin films of spin liquid candidate TbInO₃ arXiv:2603.14545, 2026 (*under review - Physical Review X*)
12. D.L. Buterakos, **S.S. Kalantre**, J. Ziegler, J.M. Taylor, J.P. Zwolak, QDFlow: A Python package for physics simulations of quantum dot devices, *SciPost Phys. Codebases*, 65 (2026)
11. M. Pendharkar, S.J. Tran, G. Zaborski Jr, J. Finney, A.L. Sharpe, R.V. Kamat, **S.S. Kalantre**, et. al. and Marc A. Kastner, Andrew J. Mannix, David Goldhaber-Gordon, Torsional force microscopy of van der Waals moirés and atomic lattices, *Proceedings of the National Academy of Sciences* 121 (10), e2314083121, 2024
10. J. Ziegler, T. McJunkin, E.S. Joseph, **S.S. Kalantre**, B. Harpt, D.E. Savage, et. al. and M.A. Eriksson, Jacob M. Taylor, Justyna P Zwolak, Toward robust autotuning of noisy quantum dot devices, *Physical Review Applied* 17 (2), 024069, 2022
9. B.J. Weber, **S.S. Kalantre**, T. McJunkin, J.M. Taylor, J.P. Zwolak, Theoretical bounds on data requirements for the ray-based classification, *SN Computer Science* 3 (1), 57, 2022
8. C.J. Trimble, M.T. Wei, N.F.Q. Yuan, **S.S. Kalantre** et. al. and J.R. Williams, Josephson Detection of Time Reversal Symmetry Breaking Superconductivity in SnTe Nanowires, *Nature Partner Journal (npj), Quantum Materials* 6 (1), 1-6, 2021
7. **S.S. Kalantre**, F. Yu, M.T. Wei, et. al. and J.R. Williams, Anomalous Phase Dynamics of Driven Graphene Josephson junctions, *Physical Review Research* 2 (2), 023093, 2020
6. J.P. Zwolak, T. McJunkin, **S.S. Kalantre**, J.P. Dodson, E.R. MacQuarrie, et. al. and M.A. Eriksson, J.M. Taylor, Autotuning of Double-Dot Devices In Situ with Machine Learning, *Physical Review Applied* 13 (3), 034075, 2020
5. J.P. Zwolak, T. McJunkin, **S.S. Kalantre**, J.P. Dodson, E.R. MacQuarrie, et. al. and M.A. Eriksson, J.M. Taylor, Ray-based framework for state identification in quantum dot devices, *PRX Quantum* 2 (2), 020335, 2020
4. J.P. Zwolak, **S.S. Kalantre**, T. McJunkin, B.J. Weber, J.M. Taylor, Ray-based classification framework for high-dimensional data, arXiv preprint arXiv:2010.00500, Proceedings of the Machine Learning and the Physical Sciences Workshop at *NeurIPS 2020*, Vancouver, Canada
3. **S.S. Kalantre**, J.P. Zwolak, et. al. and J.M. Taylor, Machine learning techniques for state recognition and auto-tuning in quantum dots, *Nature Partner Journal (npj), Quantum Information* 5 (1), 1-10, 2019
2. P. Sriram, **S.S. Kalantre**, K. Gharavi, J. Baugh, B. Muralidharan, Supercurrent interference in semiconductor nanowire Josephson junctions, *Physical Review B* 100 (15), 155431, 2019
1. J.P. Zwolak, **S.S. Kalantre**, X. Wu, S. Ragole, J.M. Taylor, QFlow lite dataset: A machine-learning approach to the charge states in quantum dot experiments, *PloS one* 13 (10), e0205844, 2018

INVITED CONFERENCE PRESENTATIONS

1. Microwave resonance in thin films of spin liquid candidate TbInO_3 , 7th International Workshop on Complex Oxides, Cargese, France, March 2024

CONTRIBUTED CONFERENCE PRESENTATIONS

1. Hybrid superconducting circuit architecture to probe van der Waals heterostructures, APS March Meeting, Denver, CO, March 2026
2. Microwave resonance probes of frustrated magnets, Capri School on Physics of Superconducting Devices, Capri, Italy, April 2025
3. Flip-chip spectroscopy of frustrated magnet $\text{Gd}_2\text{Ti}_2\text{O}_7$, MURI program on Interfacial Superconductivity, Annual Review, Stanford University, Dec 2024
4. Low temperature magneto-transport of epitaxial samarium hexaboride films, APS March Meeting, March 2024
5. Low temperature magnetotransport of SmB_6 films, MURI program on Interfacial Superconductivity, Annual Review, Harvard University, Dec 2023
6. Electron spin resonance measurements of spin liquid candidate TbInO_3 with superconducting resonators, APS March Meeting, March 2023
7. Electron spin resonance measurements of spin liquid candidate TbInO_3 with superconducting resonators, MURI program on Interfacial Superconductivity, Annual Review, Harvard University, Dec 2022
8. Electron Spin Resonance near the Metal Insulator Transition in Phosphorous doped Si, APS March Meeting, APS March Meeting, March 2022
9. Anomalous Phase Dynamics of graphene Josephson Junctions, APS March Meeting, APS March Meeting, March 2020
10. In-situ machine learning assisted auto-tuning of quantum dot devices, NISQ, UMD, June 2019
11. Machine Learning Techniques for quantum dots, JQI-QuICS-CMTC seminar, UMD, Dec 2018
12. Non-linear dynamics in graphene Josephson junctions, APS MAS, UMD, Nov 2018
13. ML and Quantum Dots, APS March Meeting, Los Angeles, CA, March 2018
14. ML and Quantum Dots, Quantum Seminar, NIST and UMD, July 2017

RESEARCH

Geballe Lab for Advanced Materials, Stanford University

June 2022 - Present

Advisor: Prof. David Goldhaber-Gordon

- ❖ Engineered a NbTiN superconducting resonator platform with flip-chip sample integration for spin resonance characterization of novel frustrated magnets and quantum spin liquid candidates.
- ❖ Designed and validated a cQED microwave architecture for superfluid stiffness extraction in van der Waals superconductors, demonstrating GHz-frequency materials characterization at mK temperatures.
- ❖ Quantified topological transport signatures in SmB_6 thin films and $\text{Eu}_3\text{In}_2\text{As}_4$ nanowires, building a magneto-transport characterization pipeline for topological material candidates.

Quantum Materials Lab (QMDLab), University of Maryland

August 2018 - June 2022

Advisor: Prof. James R. Williams

- ❖ Engineered superconducting interfaces in topological materials and van der Waals heterostructures through multi-modal electrical characterization, combining quasi-DC transport and microwave Shapiro response measurements.

- ❖ Built an analytical modeling framework for Josephson junction dynamics under microwave drive, leveraging non-linear oscillator theory to extract device parameters and predict Shapiro step signatures.

National Institute for Standards and Technology - Gaithersburg May 2017 - July 2017
Advisor: Prof. Jacob M. Taylor

- ❖ Developed CNN-based classification models for automated charge state recognition in semiconductor quantum dots, directly enabling downstream closed-loop device control.
- ❖ Engineered one of the first auto-tuning algorithms for quantum dots, autonomously driving an in-situ device to the single-electron regime — a first-of-its-kind demonstration in the field.

Indian Institute of Technology, Bombay August 2016 - May 2018
Advisor: Prof. Suddhasatta Mahapatra

- ❖ Nano-fabrication of nano-scale devices in semiconductor heterostructures (SiGe)
- ❖ Worked on a theoretical problem of Electron Dipole Spin Resonance (EDSR) selectivity of Phosphorous donor qubits in micromagnet fields

Institute for Science and Technology (IST) - Austria, Vienna May 2016 - July 2016
Advisor: Dr. Giorgos Katsaros

- ❖ Studied theoretical aspects of transport in quantum dot and nanowire systems in semi-conductors
- ❖ Worked on conductance measurements to observe Coulomb blockade and diamonds in p-type transistors at 4 K cooled with liquid He

Indian Institute of Technology, Bombay August 2016 - May 2018
Advisor: Prof. Bhaskaran Muralidharan

- ❖ Non-equilibrium Green's function (NEGF) formalism to model electron transport in superconductor-semiconductor systems
- ❖ Determined the effects of angular momentum subbands and Andreev Bound states to the current-phase relationship in nanowires connected to superconducting contacts

HONORS AND AWARDS

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| ❖ Knight-Hennessy Fellowship (\$60,000/yr)
For visionary & collaborative leaders addressing complex real-world challenges at Stanford | 2023-2026 |
| ❖ Lanczos Graduate Fellowship
Awarded to top incoming 2-3 students (\$36,000/yr) at Joint Quantum Institute, UMD | 2018-2020 |
| ❖ Institute Silver Medal for graduating at the top of Engineering Physics program, IIT Bombay | 2018 |
| ❖ S. N. Bose Scholarship (\$2000) awarded to top 50 students from India in STEM | 2017 |
| ❖ Institute Academic Award and R.P. Singh Memorial Award, IIT Bombay | 2015-2017 |
| ❖ ISTernship at IST Austria (\$2500) awarded to around 40 students worldwide | 2016 |
| ❖ National Initiative on Undergraduate Science (NIUS) Program in Physics | 2014-2015 |
| ❖ Kishore Vigyan Protsahan Yojana (KVPY), financial award for basic science high-school students | 2013 |
| ❖ 2 Gold and 1 Silver medals at International Olympiads on Astronomy and Astrophysics | 2011-2013 |
| ❖ National Talent Search Scholarship (NTSE) in science and math for high school students | 2010 |

TEACHING EXPERIENCE

Teaching Assistant, Stanford University 2023-2025

- ❖ TA for undergraduate electricity and magnetism and quantum mechanics II
- ❖ Mentored student teams in the design of a nuclear quadrupole resonance setup & cryogenic probes for advanced undergraduate physics lab

Teaching Assistant, IIT Bombay

2015-2018

- ❖ TA for a range of physics & mathematics undergraduate courses on electricity and magnetism, introductory quantum physics and partial-differential equations; on two occasions was also the head TA responsible for course logistics and personnel management

LEADERSHIP EXPERIENCE

Organizational Volunteer and Board Member, USAAAO

2018-Present

- ❖ Designed astronomy and astrophysics problems at the high-school level, and assisted in coaching the US national team to the International Olympiad on Astronomy and Astrophysics (IOAA)

General Secretary, Department of Physics, IIT Bombay

2017-2018

- ❖ Elected as the top student representative of the physics department's bachelor and masters students who communicated between students, faculty and other representatives at the institute level; work led to curriculum changes and bridged the gap in student-faculty communication in the department

Institute Student Mentor, IIT Bombay

2017-2018

- ❖ Senior mentor to a wing of 12 freshmen to assist them in settling to life at a university

Academic Resource Person, International Physics Olympiad (IPhO), Mumbai

2015

- ❖ Exam grader; worked in a team of 2 in development of IPho-rum — a browser based application for tasks such as voting, translation upload and feedback submission among ≈ 100 users

MISCELLANEOUS INTERESTS

- ❖ Science popularization at IIT Bombay at the undergraduate-level, gave talks on varied topics such as Celestial mechanics and N-body problem, Cosmology and Quantum Computing
- ❖ Violin and classical music
- ❖ Won a Bronze medal in Astronomy hackathon as part of InterIIT Tech Meet
- ❖ Secured first place with a team of five in *Online Physics Brawl 2016*
- ❖ Teaching volunteer at NGOs for underprivileged children in Powai, Mumbai